

are no downloads nor installations. The user simply logs in and accesses information. The mobile version of Facebook is an ideal example of this. It mimics the functionality of a native app but is still a website.

Advantages of PWAs

If you're still not convinced, here are some more benefits of PWAs.

- Content on a PWA is available offline as well. Unlike websites, where no content is displayed when not connected to the Internet (or if connected to a poor network), PWAs display content that users can still engage with.
- Unlike apps, PWAs don't have to be updated whenever developers roll out a new feature, bug fix or other update.
- They are optimised for performance, with the results being reflected in retention rates, customer experience, and conversions.
- No submission to app stores, i.e., no installations are needed.
- PWAs are push message-ready, and more.

Progressive Web apps are the future

So, now that we know what PWAs are and their benefits, let's understand their practical applications. Some of the companies that use PWAs include:

- Twitter Lite
- Forbes
- Alibaba
- AliExpress
- Pinterest
- Flipkart
- BookMyShow

Twitter Lite

Two years after the concept of progressive Web applications was introduced, Twitter launched its version Twitter Lite in 2017. Understanding that most of its users were accessing the portal from their phones, it ensured their experience on the mobile site was optimum. After the introduction of Lite,

Twitter reported the following numbers:

- A 75 per cent hike in the number of tweets sent
- A 65 per cent hike in pages per session
- A 20 per cent decrease in the website's bounce rates

Forbes

The case was very similar with Forbes. With the tech implementation, it reported the following:

- A 100 per cent hike in engagement rates
- A 6x hike in article completion
- A 20 per cent hike in page impressions

AliExpress

After the conversion of its mobile site into a PWA, AliExpress witnessed:

- A 74 per cent hike in the time spent during each session
- A 2x hike in new page visits
- A whopping 104 per cent hike in new user conversions

Other significant statistics

- Pinterest shared that the time spent by users on its PWA went up by 40 per cent.
- The PWA of BookMyShow was smaller than its native Android app and its iOS app by 54x and 108x, respectively.
- Flipkart reported that the amount of data used reduced by 3x after converting its mobile site to a PWA.

The basic guidelines to building PWAs

We are not going to show you how to code here since we're still at the very foundational levels of understanding PWAs. So, let's stick to the guidelines that make PWAs what they are. When you decide to build one, ensure that it has the following features.

Progressive: Your PWA should work on any device and operating system. It should also leverage the full potential of the device's hardware and other technical features.

Discoverable: Because they are not apps but websites, PWAs should be easily discoverable. This means that their SEO game must be on point, so users can immediately find them on search results and start using them.

Responsive: The site should offer a seamless Web experience across all devices.

App-like: It should look, feel, and function like a native app. This means that it should be developed on a shell model and involve minimal refreshes.

Stability even during poor connectivity: The site should work fine even in areas with poor network connectivity.

Installation-ready: If users want, they should be able to install the PWA on their browser's homepage.

Up-to-date: New content should be seamlessly pushed into the PWA when the user is accessing the site.

Besides these aspects, the PWA should be, by default, safe to use. This is because it's a website that offers a personalised and intimate user experiences. When fragile data transmission happens, the site should be safe enough to prevent middleman attacks and other threats.

So, what do you think about PWAs? An interesting concept, right? For both developers and users, they offer innumerable benefits like we just saw. With more companies opting to develop PWAs, you must offer your customers a similar experience too.

I am sure that you will now have a better understanding of the concept. I will be thrilled if you decide to implement this technology after reading this article.  **END**

 By: Pete McCain

The author is a technology startup enthusiast associated with AppVelocity, which is one of the best app developers in Toronto. He has collaborated with more than 50 entrepreneurs over the last decade, to maximise growth and contribute to technical excellence.